

Aquatech Bénin

Mobile App for Fresh Fish Sales · Digitalising Fish Farming · Benin

Research · Personas · User Flows · Wireframes · User Testing

01 P R O B L E M

Context & Challenge

Aquatech Bénin is a startup formalised in late 2023 by Sègla Grégoire Avoce. The mobile app connects fish farmers and consumers (households, restaurants) for direct fresh fish sales. Centralised model: Aquatech buys directly from partner producers, sets a uniform price, and resells via the platform. v2 deployed on Android Play Store with integrated payment, account-free navigation, and future-date reservation.

Core Problem Statement

The app has been developed and deployed, but critical UX friction blocks purchase completion: mandatory account creation before ordering, a quantity/price mismatch preventing users from calculating their total, and legally unprotected partner producers. Zero significant revenue at this stage.

Identified UX Problems

01 Mandatory Login Friction

The 'connection required' message when ordering blocks the user before the purchase act. Account creation is requested too early — before the user has made a purchase decision.

02 Quantity/Price Mismatch

Products are displayed between 300–500 grams but the price is calculated per kilogram only. Users cannot freely choose quantity or see the total price dynamically.

03 Legally Fragile Producer Partnerships

Most partner fish farmers are not formally registered. In the event of a health incident or withdrawal, accountability is unclear and Aquatech is exposed.

04 Dependency on State Authorisations

The team spent over a year waiting for state authority approvals rather than testing the market directly — delaying traction and risking displacement by a state-owned alternative.

Methodology

Research combined coaching sessions including a live test of the v2 application (coach Alimatou Diagne), competitive mapping of 3 actor types, and persona development from real user profiles identified through direct testing.

UX Competitive Analysis

UX Criterion	Fish Shops	Informal Resellers	Generic Platforms	Aquatech
Locally Farmed Fresh Fish	Partial	✓	✗	✓✓ Differentiator
Online Ordering	✗	✗	✓✓	✓✓ v2 deployed
Uniform Transparent Pricing	✗	✗	Partial	✓✓ Centralised model
Account-Free Navigation	✗	✗	Partial	✓✓ v2 integrated
Home Delivery	✗	Partial	✓✓	✓ Delivery partnerships
Producer Statistics Dashboard	✗	✗	✗	✓✓ Subscription
Future-Date Reservation	✗	✗	Partial	✓✓ Key feature

✓✓ = Very good · ✓ = Present · ✗ = Absent

User Personas

Solange B. — Household Consumer Age: 36 · Mother, Cotonou · buys fish weekly at the market Goal: Find fresh quality fish delivered home, without going to the fishmonger Frustration: Imported frozen fish doesn't taste the same. She doesn't know local fish farmers. UX Need: Browse without account, choose quantity freely and see total price, pay online, reliable delivery	Prosper A. — Partner Fish Farmer Role: Tilapia farmer, peri-urban Cotonou · 2 ponds Goal: Sell production more easily, have regular outlets, track sales Frustration: Prices vary by reseller. Never knows in advance how much he'll sell in a month. UX Need: Simple dashboard for sales stats, receive advance orders, guaranteed and stable price
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Key UX Insights

Insight #1 — Account creation must come after, not before, the purchase decision

Blocking the user with 'connection required' before completing a purchase is the single highest-friction moment in the app. The account must be requested at checkout, after the order is built — never before.

Insight #2 — Free quantity selection with dynamic pricing is non-negotiable

A user who wants 500g of tilapia but sees only a per-kg price cannot make a purchase decision. The selector must show grams + calculated total in real time: '500g = 1,500 FCFA'.

Insight #3 — Cart preservation across login is a trust requirement

When a user is forced to log in and loses their cart, they abandon the session and often don't return. Cart state must be persisted in local session so login never resets the purchase journey.

Insight #4 — Producer contract formalisation is a protective priority

Informally contracted producers expose Aquatech to unmanageable legal and health risk. Visualising which producers are formal vs. informal in the admin dashboard enables systematic prioritisation of formalisation.

User Flows — 3 Critical Journeys

Friction points were identified consistently at three moments: mandatory login before ordering, quantity/price calculation, and cart loss on login.

Journey 1 — Household/Restaurant (Buyer) Discovery (social/word of mouth) → Browse products (no account) → Choose quantity & product → Add to cart → Create account & pay → Delivery & feedback

Journey 2 — Partner Fish Farmer Convention signing → Production announcement → Aquatech buys stock → Dashboard monitoring → Sales statistics → Contract renewal

Journey 3 — Aquatech Operator Stock reception → Product listing → Order management → Delivery coordination → Invoicing & reporting → Admin dashboard analysis

Design Decisions

Based on research and live v2 testing, the following UX corrections were designed for the product page, the cart flow, and the admin dashboard for producer contract management.

Hifi Prototypes

<https://zadya22.github.io/AquatechBenin/>

User Testing Protocol

Participants: 3 Android households + 1 fish farmer + 1 internal operator

Tool: Real Aquatech v2 application

Duration: 30 min per session

#	Task	Focus Area
T1	Find fresh tilapia, choose 750g, verify total price displays correctly, add to cart without creating an account.	Quantity selection & account-free browsing
T2	Order 500g of catfish. At checkout, create an account. Verify your cart is preserved after login.	Cart persistence across login
T3	You are a partner fish farmer. Check your weekly sales stats and see which products performed best.	Producer dashboard & analytics
T4	You are the Aquatech operator. Identify producers with informal contracts and mark one as priority for formalisation.	Producer contract risk management
T5	Place an order for delivery in 3 days (future-date reservation). Verify the delivery date is confirmed on screen.	Future-date reservation confirmation

Test Report — Key Findings

T1 — Quantity and Total Price Success: 2/5	User could not freely choose quantity. Price showed per kg but products were listed at 300–500g. Total not calculated automatically. Recommendation: Free gram selector with dynamic total: '500g = 1,500 FCFA'.
T2 — Cart After Login Success: 3/5	When blocked by 'connection required', the user navigated back and lost their cart. Recommendation: Persist cart in local session storage. Login must never erase the basket.
T3 — Producer Dashboard Success: 5/5	No issues. Statistics were clear and relevant. Recommendation: Keep this format. Add a weekly/monthly trend graph to visualise performance over time.
T5 — Future-Date Reservation Success: 4/5	Delivery date was selected but no visible confirmation appeared after validation. Recommendation: Display a dedicated confirmation screen: 'Your order is reserved for [date]. A delivery partner will contact you the day before'.